

Roll No.:

END SEMESTER BACK EXAMINATION APRIL 2025

Name of the Program: B.Tech

Semester: VIIIth

Name of the Course: Soft Computing

Course Code: TCS-821

Time: 3 Hours

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1	(20 Marks)	
(a)	Explain how artificial neural networks mimic the human brain in terms of structure and functionality.	CO1
(b)	Define the term "activation function" and explain its types with appropriate examples.	CO1
(c)	What are the major terminologies associated with ANN? Explain each in detail.	CO1
Q2		
(a)	Describe the Delta learning rule and explain its application in weight adjustment.	CO2
(b)	Explain the supervised, unsupervised and reinforcement learning with appropriate examples.	CO2
(c)	Explain the concept of Adaline (Adaptive Linear Neuron). Discuss the algorithm of Adaline	CO2
Q3		
(a)	Define the cardinality of a fuzzy set. How is it computed?	CO4
(b)	Discuss the limitations of a single-layer perceptron and how it can be addressed.	CO4
(c)	What is the Perceptron learning rule? How does it work?	CO3 CO3
Q4		
(a)	Define fuzzy sets. How do they differ from classical sets? Provide suitable examples.	CO5
(b)	Illustrate the application of SOM in real-world pattern recognition problems.	CO5
(c)	List and explain various types of membership functions used in fuzzy systems.	CO4
Q5		
(a)	What is defuzzification? Describe various defuzzification techniques with examples.	CO5
(b)	With suitable examples explain the various types of crossover and mutation techniques used in genetic algorithms.	CO6
(c)	Explain how fuzzy logic can handle uncertainty and vagueness in decision-making.	CO6

